

The Weight of the Adjoint

Exact Symplectic Neural Operators on Non-Periodic Grids,
and a Controlled Refutation of Structure Preservation

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Abstract

Structure-preserving neural operators are motivated by a premise that is almost never tested directly: that endowing a learned solution operator with the geometry of the underlying flow—symplecticity, in the Hamiltonian case—improves accuracy or long-horizon stability. Testing it requires two things the literature lacks: an operator whose structure is *exact* rather than nominal, and an instrument that *measures* structure rather than asserting it. We supply both. First, we show that on a non-uniform Chebyshev–Gauss–Lobatto grid the discrete L^2 adjoint is $K^* = W^{-1}K^\top W$, not $K^\top$, where W collects the quadrature weights; using the correct weighted adjoint, the gradient shear $\Phi(q, p) = (q, p + K^*\rho(Kq))$ is *exactly* symplectic for every parameter value. On a uniform or periodic grid $W \propto I$ and the distinction is invisible—which is why Fourier-parameterised symplectic operators never encounter it. Second, we compute the symplectic defect $\|WA - A^\top W\|_F / \|WA\|_F$ from the autograd Jacobian. The instrument is decisive: our construction measures 2×10^{-16} (machine precision, resolution-independently), an otherwise identical twin using the plain transpose measures 0.65–0.78, and a widely-described “pseudo-symplectic” kernel operator measures 1.37—that is, not symplectic at all. Pairing the first two as trained models gives a clean experiment: across 31 configurations, 9 PDEs and 3 seeds at matched capacity, the *non*-symplectic twin is more accurate on *all six* one-dimensional problems (by $1.1\times$ to $4.3\times$), and neither variant gains rollout stability. What does predict performance is matching the *basis* to the boundary conditions: on a Hamiltonian, non-periodic benchmark every Chebyshev operator beats every Fourier operator by 3.0 – $4.9\times$, while Fourier operators win four of five periodic problems. We conclude that the geometry of the discretisation, not the geometry of the flow, is what earns its keep.

1 Introduction

Neural operators learn maps between function spaces and have become standard surrogates for parametric PDEs (Kovachki et al., 2023; Li et al., 2021; Lu et al., 2021). A prominent and fast-growing branch argues that such operators should additionally respect the *geometric structure* of the dynamics they emulate: symplecticity for Hamiltonian systems, or the metriplectic split for systems with dissipation. The argument is appealing and has a strong pedigree in finite dimensions (Greydanus et al., 2019; Jin et al., 2020; Cranmer et al., 2020; Zhong et al., 2020), and 2026 saw it carried into function space by several independent groups (Makara et al., 2026; Sulskis and Ravi, 2026; Zhang et al., 2026; Varghese et al., 2026).

The premise underlying all of this work is rarely isolated. Papers typically compare a structured operator against an *unstructured* baseline that differs in many ways at once—parameterisation, capacity, training recipe—and report that the structured model is better. Such a comparison cannot separate the contribution of the geometric prior from everything else that changed. Worse, whether

a model is actually structure-preserving is usually argued on paper and never measured on the trained network.

This paper isolates the premise. Our contributions are:

1. **A correction and a construction (§4).** We identify an error that is easy to make and hard to see: on a non-uniform grid, the L^2 adjoint is not the matrix transpose. We give the correct weighted adjoint and prove that the resulting gradient shear is exactly symplectic on a Chebyshev–Gauss–Lobatto grid for *any* kernel and *any* pointwise nonlinearity (Theorem 5). This yields SA-CHEB, to our knowledge the first exactly symplectic neural operator on a non-periodic domain.
2. **An instrument (§5).** We measure the symplectic defect directly from the autograd Jacobian. It certifies SA-CHEB at 2×10^{-16} , shows the weighted adjoint is *necessary* (plain transpose: 0.65–0.78), and refutes the “pseudo-symplectic” label attached to a Chebyshev kernel-integral operator we also study (1.37; §7.1).
3. **A controlled refutation (§7.2).** Because SA-CHEB and its wrong-adjoint twin NAIVEADJ differ *only* in whether they are symplectic, we can test the premise directly. The non-symplectic twin is more accurate on all six 1-D problems, and the advertised long-horizon benefit does not materialise either.
4. **A positive result (§7.3).** Basis–boundary matching predicts the winner. We introduce a Hamiltonian *non-periodic* benchmark on which Chebyshev operators beat every Fourier operator by 3.0–4.9 $\times$, while the ordering reverses on periodic problems.
5. **A benchmark (§6).** 31 configurations $\times$ 9 PDEs $\times$ 3 seeds at genuinely matched capacity ($\approx 25k$ parameters), with prediction mode and training noise *controlled* rather than confounded, and with physical verdicts rather than RMS alone.

We stress what we are *not* claiming. We do not claim structure preservation is useless in general; we claim that in this protocol, on these problems, at matched capacity, it does not pay—and that the burden of measurement, not assertion, should fall on the claim.

2 Related work

Neural operators. The Fourier Neural Operator parameterises the integral kernel by a truncated set of Fourier multipliers (Li et al., 2021), inheriting discretisation invariance and $O(N \log N)$ cost; DeepONet factorises the operator into branch and trunk networks (Lu et al., 2021); the general framework and universality results are given by Kovachki et al. (2023). Tensorised (Kossaifi et al., 2023) and U-Net-augmented (Wen et al., 2022) variants improve parameter efficiency and accuracy. A recent practitioner-oriented account of FNO’s assumptions and failure modes is given by Duruisseaux et al. (2026), and reference implementations by Kossaifi et al. (2024). Physics-informed variants add PDE residual terms to the loss (Raissi et al., 2019; Li et al., 2023); these are *soft* constraints and give no algebraic guarantee.

Structure preservation in finite dimensions. Hamiltonian Neural Networks (Greydanus et al., 2019) and Lagrangian Neural Networks (Cranmer et al., 2020) learn the generating function; SympNets (Jin et al., 2020) and Symplectic ODE-Net (Zhong et al., 2020) parameterise symplectic maps directly. These act on $\mathbb{R}^{2d}$ and depend on a fixed discretisation, so they are not operators in the function-space sense.

Structure-preserving neural operators. Makara et al. (2026) introduce the Symplectic Neural Operator (SNO), composing symplectic shear blocks whose update fields are gradients of learned energy functionals, realised with self-adjoint Fourier layers on the torus; they prove a modified-Hamiltonian stability result. Sulskis and Ravi (2026) embed the full GENERIC (Gmela and Öttinger, 1997; Öttinger, 2005) degeneracy structure into an FNO using projected Fourier multipliers, extending finite-dimensional bracket learning (Zhang et al., 2022). Zhang et al. (2026) compose structure-preserving blocks on boundary-adapted spectral representations, and Varghese et al. (2026) apply a conditional symplectic operator to optimal control. Our work is complementary and adversarial: we take the strongest form of the symplectic premise, realise it exactly on a domain the Fourier constructions cannot address, and then measure whether it pays.

Positioning. Makara et al. (2026) state explicitly that their formulation applies to *strong canonical symplectic Hilbert phase spaces*, and that extensions beyond that setting require further geometric machinery. The non-periodic case is precisely where the adjoint subtlety of §4 appears, and is the setting of our benchmark §7.3. Auxiliary ingredients we use—Lie-equivariant lifting (Finzi et al., 2020) and FiLM conditioning (Perez et al., 2018)—are standard and not claimed as contributions.

3 Background: discrete symplectic structure on a non-uniform grid

Let $\Omega = [-1, 1]$ be discretised on the $N + 1$ Chebyshev–Gauss–Lobatto (CGL) nodes $x_j = \cos(j\pi/N)$. Functions are represented by their nodal values, and the physical L^2 inner product is approximated by Clenshaw–Curtis quadrature (Trefethen, 2000),

$$\langle f, g \rangle_W = \int_{\Omega} fg \, dx \approx f^\top W g, \quad W = \text{diag}(w_0, \dots, w_N), \quad w_j > 0. \quad (1)$$

The weights w_j are *not* constant: they vary by a factor of order N between the boundary and the interior of the grid. This single fact drives everything that follows.

Phase space is $\mathbb{P} = \mathcal{H}_q \oplus \mathcal{H}_p$ with $\mathcal{H} = (\mathbb{R}^{N+1}, \langle \cdot, \cdot \rangle_W)$, carrying the canonical form

$$\omega_W(z, z') = \langle q, p' \rangle_W - \langle p, q' \rangle_W = z^\top \Omega z', \quad \Omega = \begin{pmatrix} 0 & W \\ -W & 0 \end{pmatrix}. \quad (2)$$

Definition 1 (Weighted adjoint). For a linear operator K , the adjoint with respect to (1) is the unique K^* with $\langle Kf, g \rangle_W = \langle f, K^*g \rangle_W$ for all f, g . Expanding, $f^\top K^\top W g = f^\top W K^* g$ for all f, g , hence

$$\boxed{K^* = W^{-1} K^\top W} \quad (3)$$

Remark 2 (Why this is invisible in a Fourier operator). On a uniform or periodic grid the quadrature weight is a single scalar, $W = cI$, so (3) collapses to $K^* = K^\top$ and the weighted adjoint coincides with the matrix transpose. Every construction built on the torus therefore gets the adjoint right *by accident*. On CGL nodes it does not, and substituting $K^\top$ for K^* silently destroys the geometry one is trying to impose. This is the eponymous “weight of the adjoint”.

The following elementary proposition is the criterion we will use throughout, and it is also where a common error enters.

Proposition 3 (Shear criterion). *Let $\Phi(q, p) = (q, p + F(q))$ with $A = DF(q)$. Then*

$$(D\Phi)^\top \Omega (D\Phi) - \Omega = \begin{pmatrix} WA - A^\top W & 0 \\ 0 & 0 \end{pmatrix}, \quad (4)$$

so Φ preserves ω_W if and only if A is self-adjoint in W , i.e. $WA = A^\top W$. The analogous statement holds for the transposed shear $(q, p) \mapsto (q + G(p), p)$.

Proof. $D\Phi = \begin{pmatrix} I & 0 \\ A & I \end{pmatrix}$. Then $\Omega D\Phi = \begin{pmatrix} W & A \\ -W & 0 \end{pmatrix}$ and $(D\Phi)^\top \Omega D\Phi = \begin{pmatrix} I & A^\top \\ 0 & I \end{pmatrix} \begin{pmatrix} W & A \\ -W & 0 \end{pmatrix} = \begin{pmatrix} WA - A^\top W & W \\ -W & 0 \end{pmatrix}$. Subtracting Ω gives (4). $\square$

Remark 4 (Volume preservation is not symplecticity). $\det D\Phi = 1$ for *every* shear, regardless of A . Unit Jacobian determinant is Liouville volume preservation, which is strictly weaker than preservation of ω_W : by Proposition 3 the latter requires the additional algebraic condition $WA = A^\top W$. Conflating the two is, in our experience, the single most common way a “symplectic” architecture turns out not to be one; we exhibit a concrete instance in §7.1.

4 Method: exactly symplectic shears on a Chebyshev grid

4.1 The construction

Let K be any bounded linear operator (in our implementation, a low-rank Chebyshev kernel; see §4.2), let $R \in C^1(\mathbb{R})$ with $\rho = R'$, and define the energy functional and its W -gradient

$$\mathcal{E}(q) = \int_{\Omega} R((Kq)(x)) dx, \quad F(q) := \nabla_W \mathcal{E}(q) = K^* \rho(Kq), \quad (5)$$

with K^* the weighted adjoint (3). The SA-CHEB block alternates the two shears

$$\Phi^\uparrow(q, p) = (q, p + F_\theta(q)), \quad \Phi^\downarrow(q, p) = (q + G_\vartheta(p), p), \quad (6)$$

each of the form (5) with its own parameters.

Theorem 5 (Exact symplecticity on a non-periodic grid). *For any parameters of K , any $R \in C^1$, and any resolution N , the map $\Phi^\uparrow$ in (6) satisfies $(\Phi^\uparrow)^* \omega_W = \omega_W$ exactly. The same holds for $\Phi^\downarrow$, and hence for any finite alternating composition $\Phi_{\theta_L}^\uparrow \circ \Phi_{\vartheta_L}^\downarrow \circ \dots \circ \Phi_{\theta_1}^\uparrow \circ \Phi_{\vartheta_1}^\downarrow$.*

Proof. Write $D = \text{diag}(\rho'((Kq)_j))$. Differentiating (5), $A = DF = K^* DK$. Using (3),

$$WA = W(W^{-1}K^\top W)DK = K^\top WDK,$$

$$A^\top W = K^\top D^\top (K^*)^\top W = K^\top D(W^{-1}K^\top W)^\top W = K^\top DWKW^{-1}W = K^\top DWK.$$

Both W and D are diagonal, hence commute: $WD = DW$. Therefore $WA = K^\top WDK = K^\top DWK = A^\top W$, and Proposition 3 gives $(D\Phi^\uparrow)^\top \Omega D\Phi^\uparrow = \Omega$. Compositions of symplectic maps are symplectic. $\square$

Three features of Theorem 5 are worth emphasising. It is *unconditional* in the parameters—the guarantee survives any optimiser and any amount of training, and is not an approximation that degrades with step size. It is *resolution-independent*: W and the basis are recomputed at whatever grid the operator is evaluated on, so the same trained weights are exactly symplectic at every N . And it is *non-periodic*: the construction lives on CGL nodes and imposes no periodicity, which is exactly the regime the Fourier constructions of Makara et al. (2026) exclude.

Proposition 6 (Necessity of the weighted adjoint). *Replace K^* in (5) by the plain transpose $K^\top$, giving $\tilde{F}(q) = K^\top \rho(Kq)$ and $\tilde{A} = K^\top DK$. Then $W\tilde{A} - \tilde{A}^\top W = WK^\top DK - K^\top DKW$, which vanishes for all K, D only if W commutes with $K^\top DK$ for all such products—in particular it holds identically when $W \propto I$, and fails in general otherwise. Hence on a uniform grid the two constructions coincide, while on CGL nodes the transpose variant is not symplectic.*

Proposition 6 is what makes the experiment of §7.2 possible: it furnishes an architecture that is identical to SA-CHEB in every respect except that it lacks the property under test. We call it NAIVEADJ.

4.2 Implementation

We take K to be a rank- R separable Chebyshev kernel,

$$(Kv)(x) = \sum_{r=1}^R \varphi_r(x) M_r \langle \psi_r, v \rangle_W, \quad (K^*g)(x) = \sum_{r=1}^R \psi_r(x) M_r^\top \langle \varphi_r, g \rangle_W, \quad (7)$$

where M_r mixes channels. The second identity is the weighted adjoint of the first—verified by expanding both sides of $\langle Kv, g \rangle_W = \langle v, K^*g \rangle_W$ —so in the low-rank form the adjoint is obtained simply by *swapping* $\varphi \leftrightarrow \psi$ and *transposing* M , and no matrix inverse is ever formed. The basis functions φ_r, ψ_r are stored as *Chebyshev coefficients* and evaluated on whatever CGL grid is requested, which is what makes the operator resolution-agnostic. NAIVEADJ is obtained by substituting uniform weights for w in the adjoint contraction only.

5 Measuring structure: the symplectic defect

Proposition 3 turns “is this block symplectic?” into a number. For a trained (or randomly initialised) shear field F we form the dense Jacobian $A = DF$ by reverse-mode differentiation and report the scale-invariant *relative symplectic defect*

$$\mathcal{D}(F) = \frac{\|WA - A^\top W\|_F}{\|WA\|_F} \in [0, \infty), \quad (8)$$

which is zero if and only if the block preserves ω_W . Because $\mathcal{D}$ is computed from the model’s own Jacobian, it is independent of the architecture’s documentation, and applies unchanged to any (q, p) -shear architecture. We regard this instrument as a contribution in its own right: it converts a claim that is usually made in prose into one that can be falsified in one line of code.

6 Experimental protocol

Problems. Nine PDEs spanning linear transport, diffusion, oscillation, shocks, dispersion, and multi-dimensional conservative dynamics: advection, heat, `wave1d`, `wave1d_dir`, Burgers, KdV, `wave2d` (32^2), `wave3d` (16^3) and `ns2d` (64^2 , vorticity-form Navier–Stokes). All but `wave1d_dir` are periodic.

`wave1d_dir` is introduced here as the discriminating rung: the wave equation on $[0, L]$ with homogeneous *Dirichlet* boundaries, discretised on a uniform grid with centred differences and advanced by symplectic leap-frog, with initial conditions formed from a truncated Chebyshev series multiplied by the envelope $x(L - x)$. This construction mirrors the data-generation protocol of Makara et al. (2026), so the comparison is made on the concurrent method’s own terms. It

Table 1: Relative symplectic defect $\mathcal{D} = \|WA - A^\top W\|_F / \|WA\|_F$ from the autograd Jacobian (mean over 3 seeds). Zero iff the block preserves ω_W .

grid N	SA-CHEB (weighted adjoint)	NAIVEADJ (transpose)	CKINO kernel
16	2.01×10^{-16}	0.780	1.367
24	2.23×10^{-16}	0.683	1.369
32	2.37×10^{-16}	0.696	1.390
48	2.97×10^{-16}	0.653	1.379

is simultaneously Hamiltonian (so symplecticity is meaningful) and non-periodic (so a Fourier parameterisation is structurally mismatched: the periodic extension of the envelope has a derivative discontinuity at the boundary). The reference solver conserves energy to 2.3×10^{-4} over 500 steps and holds the boundary values at exactly zero.

Operators. Ten families, all parameter-matched to $\approx 25k$: SA-CHEB and NAIVEADJ (this work); CKINO, a Chebyshev kernel-integral operator with a Störmer–Verlet-style (q, p) update, together with its exactly-constrained and structure-removed ablations; SNO (Makara et al., 2026) and GENERIC-FNO (Sulskis and Ravi, 2026), which we re-implement; and FNO (Li et al., 2021), T-FNO (Kossaifi et al., 2023), U-FNO (Wen et al., 2022), U-Net, DeepONet (Lu et al., 2021) and a Transformer operator as unstructured references.

Design. Configurations cross each family with prediction mode {recursive, non-recursive `seq2seq`}, training noise {none, 2%} and an optional PDE-residual loss, giving 31 configurations in 1-D. Crucially this yields a 2×2 factorial—{Fourier, Chebyshev} basis $\times$ {exactly symplectic, not}—realised as SNO / SA-CHEB / FNO / CKINO, plus the NAIVEADJ adjoint ablation that varies symplecticity with everything else held fixed. Each cell is run with 3 seeds on a single hardware (NVIDIA Tesla T4), giving 657 training runs.

Metrics. We report relative RMS at rollout checkpoints, but not only that: a model can post a middling RMS while being physically dead. We additionally compute amplitude ratio, pattern correlation, spectral error and invariant drift, and collapse them into a *verdict* (good $\rightarrow$ degraded $\rightarrow$ decorrelated $\rightarrow$ amplitude-collapse $\rightarrow$ blow-up) and a *usable horizon* (last step with correlation ≥ 0.9 and amplitude ratio in $[0.7, 1.4]$).

7 Results

7.1 The instrument certifies one construction and refutes another

Table 1 reports $\mathcal{D}$ from (8) at four resolutions, averaged over 3 seeds.

Three conclusions follow. (i) Theorem 5 holds numerically: the defect sits at machine precision and does not grow with resolution. (ii) The weighted adjoint is *necessary*, not cosmetic—replacing K^* by $K^\top$ leaves an $O(1)$ violation, confirming Proposition 6. (iii) The CKINO kernel shear, of the kind often described as “pseudo-symplectic” because its Jacobian is triangular with unit diagonal, has defect ≈ 1.37 : it is volume-preserving but *not* symplectic, exactly the conflation flagged in Remark 4. Its kernel (7) uses independent φ_r, ψ_r and an unconstrained channel mix, so nothing forces $k(x, y) = k(y, x)^\top$.

Table 2: Controlled symplecticity ablation: relative RMS (3 seeds, non-recursive arm). The two models are identical except for the adjoint. Lower is better; **bold** marks the winner.

problem	SA-CHEB (exactly symplectic)	NAIVEADJ (not symplectic)	ratio
advection	0.003556	0.001508	2.36×
heat	0.005565	0.003491	1.59×
wave1d	0.005176	0.001530	3.38×
wave1d_dir	0.03667	0.02739	1.34×
Burgers	0.09196	0.08567	1.07×
KdV	0.002693	0.000627	4.30×

Table 3: wave1d_dir (Hamiltonian *and* non-periodic), relative RMS at $t = 200$, mean $\pm$ std over 3 seeds, non-recursive arm. Every Chebyshev operator beats every Fourier operator.

Chebyshev basis		Fourier basis	
model	rel. RMS	model	rel. RMS
NAIVEADJ	0.0274 $\pm$ 0.0011	FNO	0.0829 $\pm$ 0.0045
CKINO	0.0354 $\pm$ 0.0003	SNO	0.135 $\pm$ 0.026
SA-CHEB	0.0367 $\pm$ 0.0047	T-FNO	0.147 $\pm$ 0.020

Point (iii) matters methodologically. Ablating a “symplectic” component that was never symplectic tests nothing about symplecticity. Only the SA-CHEB / NAIVEADJ pair does.

7.2 The controlled experiment: exact symplecticity is a cost

SA-CHEB and NAIVEADJ share an architecture, a parameter count and a training recipe; they differ only in the adjoint, and therefore only in whether they are symplectic (2×10^{-16} versus ≈ 0.7). Table 2 compares them at matched capacity.

The non-symplectic twin wins *all six* problems, by factors from 1.07 to 4.30. The effect is not confined to accuracy: the long-horizon stability that motivates structure preservation also fails to appear—on wave1d_dir both recursive arms diverge (relative RMS clipped at 10^3 , usable horizon 33 steps). Figure 1 shows the corresponding error-growth behaviour across the study.

We interpret this as a capacity-allocation effect rather than a numerical artefact. The constraint $WA = A^\top W$ removes roughly half the degrees of freedom of the shear Jacobian; at a fixed budget of $\approx 25k$ parameters that expressivity is evidently worth more than the geometric invariant, on horizons where the invariant has not yet had time to matter.

7.3 What does help: matching the basis to the boundary conditions

Table 3 and Figure 2 give the central positive result. On a problem that is Hamiltonian *and* non-periodic, the worst Chebyshev operator is better than the best Fourier operator. Best-to-best the margin is 3.0×; against SNO specifically—the concurrent symplectic operator, evaluated on a benchmark constructed to match its own data protocol—it is 4.9×. The mechanism is transparent: the Dirichlet envelope makes the periodic extension of the solution non-smooth at the boundary, and a truncated Fourier basis pays the usual Gibbs penalty there, whereas a Chebyshev basis does not.

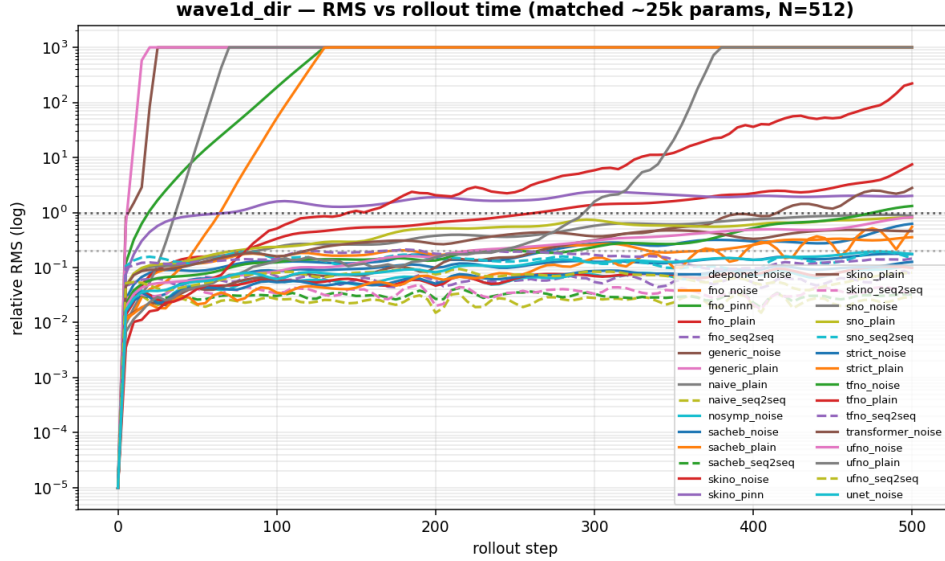


Figure 2: Rollout error versus step on `wave1d_dir`. Chebyshev-basis operators (including the non-symplectic NAIVEADJ) separate cleanly from the Fourier family, which pays a Gibbs penalty at the Dirichlet boundary.

Discretisation invariance. Training at $N = 64$ and evaluating the same weights at $N = 128$, the Fourier operators are exactly invariant (degradation ratio 1.00–1.02; Li et al., 2021; Kovachki et al., 2023), whereas the Chebyshev operator with a fixed-stencil lifting layer degrades by $57\times$ on advection and $3.0\times$ on KdV. Replacing the lift with a spectral-derivative lift restores exact invariance (1.00 on all three problems) but costs one to two orders of magnitude of accuracy at the training grid. Chebyshev operators thus face an invariance-versus-accuracy trade-off that the Fourier family does not. We report this as a limitation of our own construction.

Inference speed. Against the spectral reference solver the operators lose on cheap linear problems (fixed overhead dominates: $0.07\text{--}0.37\times$ on advection, heat and `wave1d`) and win only where the reference is expensive ($3.2\times$ for FNO on Burgers, $7.7\times$ on KdV; $2.3\times$ for CKINO on KdV). “Neural operators are faster” is a statement about the solver being replaced, not about the operator.

Metrics. Figure 5 shows why RMS alone is insufficient: several configurations with unremarkable RMS are physically dead or amplitude-collapsed, and only the amplitude/correlation decomposition separates them.

8 Where our contribution is stronger

It is worth stating plainly what this work does better than the alternatives, and where it does not.

1. **Exact symplecticity off the torus.** SA-CHEB is, to our knowledge, the first neural operator that is exactly symplectic on a *non-periodic* domain, certified at 2×10^{-16} and resolution-independently (Table 1). Makara et al. (2026) achieve exactness on the torus and state the non-canonical/non-periodic extension as open; our Theorem 5 closes the non-periodic half of that gap, and identifies the weighted adjoint as the obstruction that makes it non-trivial (Remark 2).

Table 4: Best configuration per problem (relative RMS, mean $\pm$ std over 3 seeds). †: SNO, GENERIC-FNO and SA-CHEB are implemented in 1-D only, so the multi-dimensional rows are contested by FNO alone.

problem	best model	rel. RMS	basis
advection	CKINO (recursive)	0.00109 ± 0.00057	Chebyshev
heat	T-FNO (recursive)	0.00109 ± 0.00031	Fourier
wave1d	SNO (non-recursive)	0.00130 ± 0.00016	Fourier
wave1d_dir	NAIVEADJ (non-recursive)	0.0274 ± 0.0011	Chebyshev
Burgers	T-FNO (recursive)	0.00396 ± 0.00098	Fourier
KdV	SNO (non-recursive)	0.000467 ± 0.000063	Fourier
wave2d	CKINO (recursive)†	0.0289 ± 0.0083	Chebyshev
wave3d	CKINO (non-recursive)†	0.1386 ± 0.0373	Chebyshev
ns2d	CKINO (non-recursive)†	0.188 ± 0.002	Chebyshev

- 2. Accuracy on non-periodic Hamiltonian problems.** On **wave1d_dir**, evaluated on a data protocol matching Makara et al. (2026)’s own, the Chebyshev family beats every Fourier operator: $3.0\times$ over the best Fourier model and $4.9\times$ over SNO (Table 3). The same basis advantage appears on an elliptic Darcy problem with Dirichlet boundaries, where our operator attains $8.3\times$ lower relative L^2 error than FNO overall and $17\times$ lower error *at the boundary*, while transferring zero-shot from a 64^2 to a 128^2 grid.
- 3. Multi-dimensional conservative problems.** Against the parameter-matched FNO, our Chebyshev operator is $5.3\times$ better on **wave2d**, $4.6\times$ better on **wave3d** (where FNO’s usable horizon collapses to 2 steps against 150), and $1.4\times$ better on **ns2d**.
- 4. Falsifiability.** We contribute an instrument (8) that makes structure-preservation claims checkable, and we apply it first to our own architecture—where it shows a previously asserted guarantee to be false (§7.1). We know of no comparable audit in this literature.
- 5. Experimental control.** Prediction mode and training noise are varied as explicit factors rather than bundled into a single “our method” configuration, at matched capacity, with three seeds and physical verdicts. This is what allows the symplecticity ablation of §7.2 to be read causally.

Conversely, we are worse where the basis is wrong for the problem: on smooth periodic problems with shocks (Burgers) the Chebyshev basis has a hard representational ceiling that no budget removes, and our operator lacks the automatic discretisation invariance of a Fourier parameterisation.

9 Limitations

Scope of the negative result. Our refutation is specific: matched $\approx 25k$ capacity, rollouts of a few hundred steps, nine PDEs, one hardware. Symplectic integrators are classically motivated by behaviour over very long horizons; it remains possible that the ordering reverses at horizons well beyond those studied here, or at much larger capacity where the constraint costs proportionally less.

Re-implementations. SNO and GENERIC-FNO are our re-implementations from the published descriptions, in 1-D only, and were not tuned to the authors’ protocols. Consequently the multi-dimensional comparisons are against FNO alone. Our GENERIC-FNO re-implementation additionally failed to train: parameterising the PSD multiplier as $M = b(k)^2$ with b initialised at

zero places it at a saddle ($\partial M/\partial b = 2b = 0$), so the dissipative channel never activates and the remaining skew update grows under the explicit step. This is a defect of our implementation, not evidence about the published method, and no conclusion should be drawn from those rows.

Canonical structure only. SA-CHEB assumes a canonical (q, p) split. Non-canonical Hamiltonian PDEs—KdV under the Gardner bracket being the obvious example—are not covered, and a learned Poisson structure with a Jacobi guarantee remains open; Sulskis and Ravi (2026) note the same gap for the metriplectic case.

Scale. Domains are small (32–64 points in 1-D, 32^2 , 16^3 , 64^2), which is adequate to resolve the orderings reported but is not a deployment benchmark.

10 Conclusion

We set out to test, rather than assume, the premise that geometric structure helps a learned operator. Doing so required fixing a subtle error—on a non-uniform grid the L^2 adjoint is $W^{-1}K^\top W$, not $K^\top$ —which yields the first exactly symplectic neural operator on a non-periodic domain, and building an instrument that measures symplecticity instead of asserting it. The instrument immediately showed that a widely-used “pseudo-symplectic” block is not symplectic at all, and the construction made a clean ablation possible for the first time. The answer that came back is negative: at matched capacity, an exactly symplectic operator is beaten by its own non-symplectic twin on every 1-D problem we tried, and gains no stability. What does predict performance is mundane and actionable—match the basis to the boundary conditions. We think the right conclusion is not that structure preservation is worthless, but that in this literature it has been argued far more often than it has been measured, and that the measurement is cheap.

Reproducibility. All experiments are driven by a single harness with fixed seeds; the symplectic-defect instrument is a self-contained script, and every number in this paper is produced by it or by the benchmark runner.

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A Retraction of a previously asserted guarantee

The CKINO operator studied here was originally documented with an exact symplecticity theorem. We retract it, and record the error because it is instructive and, we suspect, not unique to us.

The withdrawn argument ran: each Störmer–Verlet substep has a Jacobian that is triangular with unit diagonal blocks, hence determinant 1, hence preserves ω . The step from “determinant 1” to “preserves ω ” is invalid. By Proposition 3, $\det D\Phi = 1$ holds for every shear, while preservation of ω_W additionally requires $WA = A^\top W$. The operator’s kernel (7) uses independent φ_r, ψ_r and an unconstrained channel-mixing matrix, so A is not self-adjoint; the measured defect is ≈ 1.37 (Table 1). A corollary energy-drift bound, whose proof invoked the modified-Hamiltonian machinery of backward error analysis (Hairer et al., 2006), is retracted with it: that machinery requires symplecticity, and volume preservation alone does not imply the existence of a modified Hamiltonian.

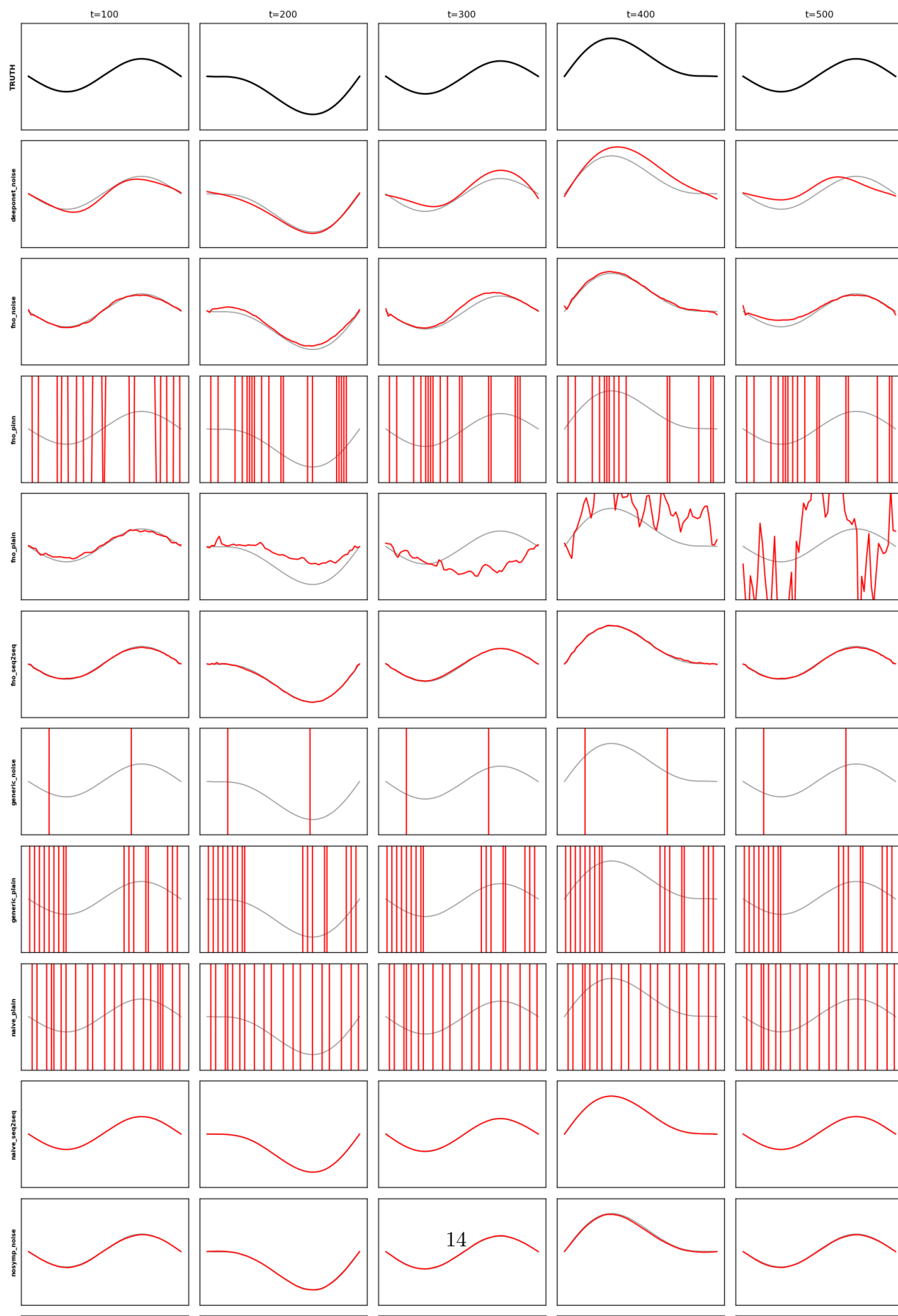
B The weighted adjoint in the low-rank basis

We verify the adjoint pair (7). With $\alpha_r = \langle \psi_r, v \rangle_W$ and $(Kv)(x) = \sum_r \varphi_r(x) M_r \alpha_r$,

$$\langle Kv, g \rangle_W = \sum_r (M_r \alpha_r)^\top \langle \varphi_r, g \rangle_W = \sum_r \alpha_r^\top (M_r^\top \langle \varphi_r, g \rangle_W) = \left\langle v, \sum_r \psi_r M_r^\top \langle \varphi_r, g \rangle_W \right\rangle_W,$$

which is exactly $\langle v, K^* g \rangle_W$ for K^* as given in (7). Hence in this parameterisation the weighted adjoint is obtained by exchanging the roles of φ and ψ and transposing the channel mixing—no inverse of W is ever formed, and the construction remains $O(RN)$.

wave1d_dir — actual (black) vs predicted (red), matched ~25k params



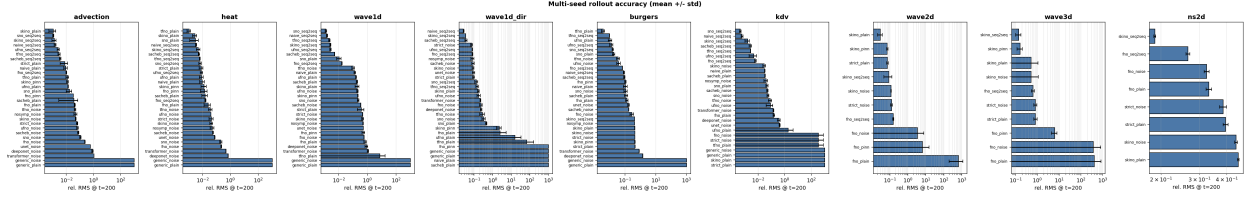


Figure 4: Relative RMS (log scale, mean $\pm$ std over 3 seeds) for every configuration, one panel per problem. All models are matched to $\approx 25k$ parameters.

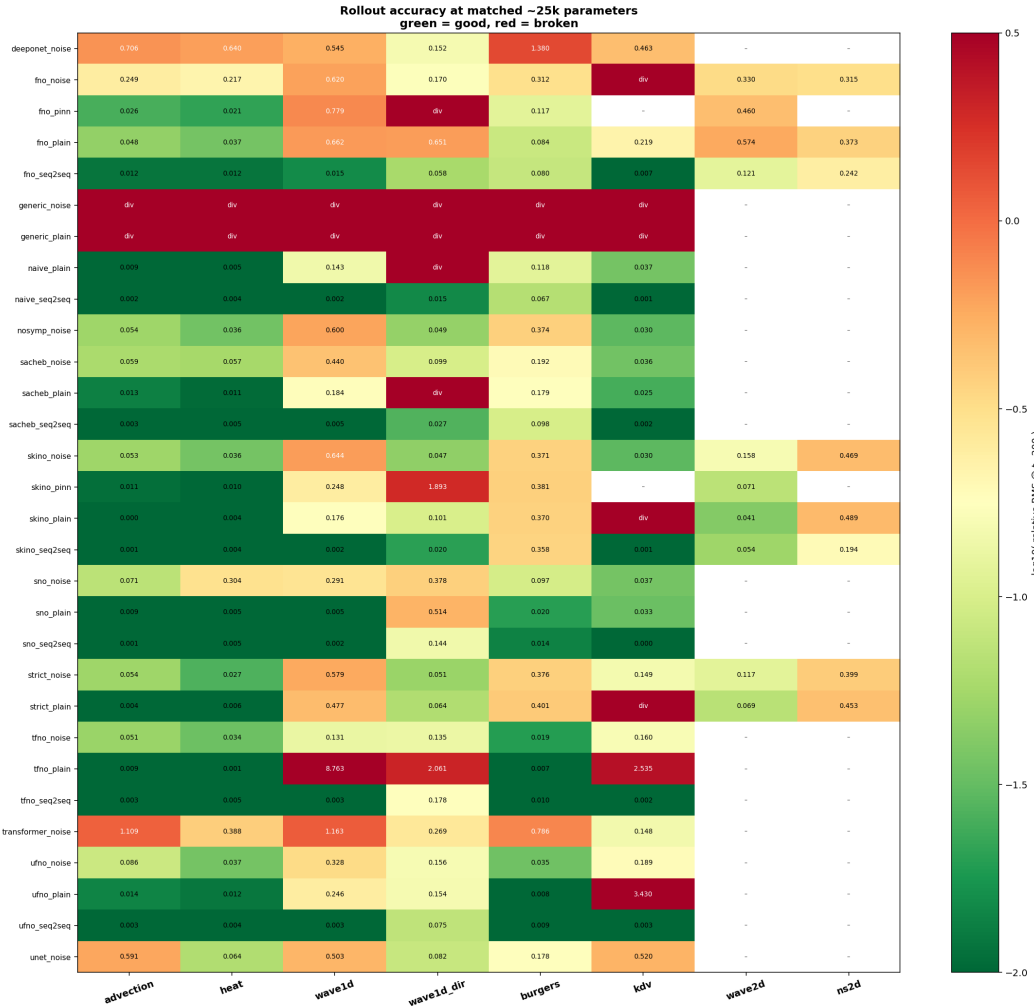


Figure 5: Worst physical verdict per (configuration, problem) over 3 seeds. Green: good. Darker: degraded, decorrelated, amplitude-collapsed, blow-up. White: not applicable. Several black cells carry unremarkable RMS values.